

Emilio Gullón López

Software Engineer · Data Science & Machine Learning
University of Granada (UGR) · Specialization in AI & Data Science

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PROFESSIONAL SUMMARY

Computer Science graduate from the **University of Granada (UGR)**, specializing in **Artificial Intelligence and Data Science**. My technical value proposition blends software engineering rigor (modular design patterns, robust data structures, and concurrency) with applied research in machine learning, multimodal time-series modeling, and deep neural architectures. Driven by scientific reproducibility, building leakage-free data pipelines, and deploying low-latency models for production and Edge AI environments.

TECHNICAL SKILLS & TECHNOLOGY STACK

Programming Languages & Databases

Python (Advanced), **SQL** (PostgreSQL, MySQL, Oracle SQL), **C++**, **Java**, **JavaScript**, **C#**.

Focus: Strict type hinting, PEP8 standards, complex query optimization, B-Tree/GIN indexing, and ACID transactional integrity.

AI, Deep Learning & Data Science

PyTorch, **Scikit-learn**, **Optuna**, **OpenCV**, Pandas, NumPy, SciPy, Keras, Fastai, Matplotlib, Seaborn.

Areas: Multimodal Time-Series, Recurrent Architectures (Custom GRU, xLSTM), Transformers, Computer Vision, and Bayesian Optimization.

Software Engineering & Web Development

Django, **RESTful APIs**, **MVC** Architecture, **SOLID** Principles, Modular Design, Unit Testing, Clean Refactoring, Concurrent and Asynchronous Programming.

Infrastructure, DevOps & MLOps

Git / GitHub, **Docker**, **Linux** (Ubuntu, Debian), WSL, CI/CD, Jupyter Notebooks, Python Package Distribution (pip/wheel).

FEATURED ENGINEERING PROJECT

B.S. Thesis: Predictive Fatigue Analysis with Multimodal Biosignals

University of Granada (UGR) · Sept. 2025 – 2026

Final Grade: 9.8 / 10

Web: [emiliogullon.github.io/tfg-fatigueset-web](#)

Technologies: Python PyTorch xLSTM Custom GRU Transformers Optuna Edge AI

- Pipeline Architecture & Modular Library (fatigueset-lib):** Designed and implemented a decoupled Python/PyTorch package for harmonic synchronization (1 Hz to 256 Hz) and biomarker extraction across multimodal time-series, enforcing strict subject-wise partitioning that eradicated 100% of data leakage.
- Sequential Modeling & Edge AI Optimization:** Conducted benchmarking and Bayesian hyperparameter tuning (**Optuna**) across Custom GRU, xLSTM, and Transformers; formulated multi-objective **Pareto-frontier** optimization for ultra-low-latency on-device inference and local data privacy (*EU AI Act*).

TECHNICAL PROJECTS & PRACTICAL EXPERIENCE

Biomedical Semantic Segmentation in Computer Vision

Computer Vision Specialization · University of Granada

Computer Vision

Technologies: PyTorch Fastai OpenCV U-Net DRIVE Dataset

- Developed an end-to-end retinal vascular segmentation pipeline (DRIVE dataset) training deep convolutional **U-Net** networks with PyTorch and Fastai.
- Applied adaptive local contrast enhancement (**CLAHE** in $L^*a^*b^*$) and compound loss functions (**Dice Loss** and IoU) mitigating severe class imbalance in microvasculature.

Modular Web Platform & Transactional Persistence

Software Engineering Track · University of Granada

Software Engineering

Technologies: Django Python PostgreSQL REST APIs HTML5/CSS3

- Built a modular web application following the **MVC** architectural pattern in Django, with relational data modeling in PostgreSQL and RESTful API endpoints.
- Implemented role-based access control (RBAC), unit testing suites, and structural decoupling adhering to SOLID design principles.

Metaheuristic Optimization & Systems Simulation

Advanced Algorithms & Modeling · University of Granada

Algorithms & Simulation

Technologies: C++ Python Discrete Simulation Genetic Algorithms Tabu Search

- Implemented and evaluated metaheuristic algorithms (Genetic Algorithms, Tabu Search, Simulated Annealing) for combinatorial optimization on NP-hard problems.
- Designed computational models and discrete-event simulations for stochastic systems in C++/Python, analyzing waiting times, queues, and system convergence.

EDUCATION & ACADEMIC BACKGROUND

B.S. in Computer Science and Engineering

Sept. 2021 – 2025/2026

University of Granada (UGR) · Granada, Spain

240 ECTS Completed

- Specialization in Artificial Intelligence and Data Science:** Advanced coursework in Machine Learning, Computer Vision, Data Mining, Intelligent Systems, Metaheuristics, and Systems Simulation.
- Computer Science Core & Software Engineering Foundations:** Advanced algorithms, complex data structures, computer architecture, object-oriented design, operating systems, networks, and concurrent programming.

LANGUAGES & ENGINEERING METHODOLOGY

Language Proficiency

Spanish: Native (fluent oral and written communication).

English: Professional proficiency in technical reading and writing; actively preparing for B2 certification.

Portuguese: Basic working proficiency.

Engineering Rigor & Methodology

Clean Code: Modular, testable, and maintainable software under SOLID design principles.

Scientific Rigor: Rigorous experimental tracking, versioning, and reproducibility.

Production Mindset: Focus on computational efficiency, overfitting mitigation, and observability.